Aric Maji

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.79
Delhi Public School Ruby Park Kolkata	Class XII (CBSE)	2024	95.60%
Delhi Public School Ruby Park Kolkata	Class X (CBSE)	2022	97.80%

SCHOLASTIC ACHIEVEMENTS

- Attained a Peak Rating of **1733 (Expert)** on Codeforces, solved **400+** problems across different platforms. (Handle)
- Achieved **Global Rank 145** and **India Rank 27** in **Codeforces Round 1108 (Div 2)** among 27,000+ participants.
- Secured a funding of **20,000** from the **BUILD Initiative** by the IITH Incubator for an innovative project idea.
- Shortlisted for the **final stage** of the **Tredence Infinity Hackathon 2025** among **1000+** participating teams.
- Secured an **A+ grade** in Software Development Fundamentals and DBMS, and an **A grade** in core courses including Introduction to Programming, Discrete Mathematics, and Introduction to Computing.

PROJECTS

Key Management System (KMS) | *Go, React, TypeScript, SQLite, AWS, OAuth* | **Github** **Live Demo**

- Built a production ready, official **key and room access** management platform for **IIT Hyderabad** with a modular **Go** (Gin) backend, **SQLite**, and **20+ REST endpoints** for access management and audit logging.
- Implemented a Role-Based Access Control (**RBAC**) system using Google **OAuth** and stateless HTTP-only **JWTs**, enabling secure key provisioning and real-time checkout workflows.
- Developed and deployed the responsive **React** application on **AWS EC2** with **Nginx** reverse proxy and **Systemd**.

RISC-V Pipeline Simulator | *C++, TypeScript, Bash* | **Github** - Simulator GUI

- Built a **cycle-accurate 5-stage RISC-V pipeline simulator** supporting the RV64I, F, and D instruction set extensions, with configurable execution modes for hazard detection, forwarding and branch prediction.
- Reduced **control hazard penalties by 50%** through early branch resolution, static/dynamic branch prediction, and built a configurable cache simulator.
- Developed a **VS Code extension** using DAP with pipeline visualization, snapshot debugging, and automated testing for correctness and performance.

KRC Library Application | *Go (net/http), Flutter, PostgreSQL* | **Github**

- Developed the discussion room booking and administrative approval workflow for the official IIT Hyderabad Library application using a **Go** (net/http) backend, **PostgreSQL**, and a **Flutter** client.
- Built a concurrent reservation system using **PostgreSQL transaction locks** to prevent race conditions, with **Provider** based state management, **Dio** API integration, and structured client-side validation.

SplitEase: Expense Sharing Platform | *Go, Flutter, MongoDB, JWT* | **Github** **Live Demo**

- Built a full stack expense sharing platform using **Go**, **Flutter** Web, and **MongoDB** with a layered **REST API** secured using stateless **JWT** authentication.
- Implemented a **greedy minimum cash flow algorithm** to optimize debt settlements across equal, proportional, and uneven expense splits.

xv6 Operating System Extension | *C, xv6, OS Arch* | **Github**

- Extended the xv6 OS by building a syscall-aware Multilevel Feedback Queue (**SC-MLFQ**) scheduler with dynamic

time quanta, priority boosting, and syscall-based demotion policies.
- Implemented a **demand-paging virtual memory manager** with lazy allocation, a priority-aware Clock page replacement algorithm, and thread-safe kernel synchronization using fine-grained locking.
- Built a **virtual storage subsystem** featuring priority-aware disk scheduling (FCFS/SSTF) and a Software **RAID engine** supporting RAID 0, 1, and 5 with XOR-based parity reconstruction.

Arbitrary Precision Arithmetic Library | *Java, Maven, Docker, JUnit* | **Github** **Docker Hub**

- Built a **Java library** and CLI utility implementing arbitrary precision arithmetic beyond native integer limits.
- Implemented **string-based arithmetic** algorithms with configurable decimal precision and validated correctness using comprehensive **JUnit test suites**, and packaged the project with **Maven** & **Docker**.

SKILLS

Programming Languages: C, C++, Java, Go, Python, JavaScript, TypeScript, OCaml, SQL, Bash, Assembly (RISC-V).

Frameworks: React.js, Next.js, Express.js, Flutter, Gin, Flask, FastAPI.

Databases: PostgreSQL, MongoDB, MySQL, SQLite.

Dev Tools: Git, Docker, AWS, Nginx, Linux, Maven, Make, JWT, OAuth.

RELEVANT COURSES

Computer Science: DSA, OS, DBMS, Computer Architecture, Compilers, Computation Theory, Software Dev.

Mathematics: Probability& Random Variables, Linear Algebra, Statistics, Calculus, Transform Techniques, Discrete Math.

POSITIONS OF RESPONSIBILITY

Overall Head at **Lambda**, **The Software Development Club** of **IIT Hyderabad**

- Leading a team of 20+ developers to build institute wide open-source software initiatives and host technical workshops.

Dev Core at **Lambda**, **The Software Development Club** of **IIT Hyderabad**

Software Development Coordinator at **Tinkerers' Lab**, **IIT Hyderabad**

Teaching Assistant for **Discrete Mathematics** (CS1010)

Web Coordinator at **Elan & nVision**, **The Techno-Cultural Fest** of **IIT Hyderabad**

EXTRACURRICULAR

Won 2nd and 3rd place in the Milan Inter-Hostel Hackathon at IITH.

Active participant in various online CTF cybersecurity competitions.